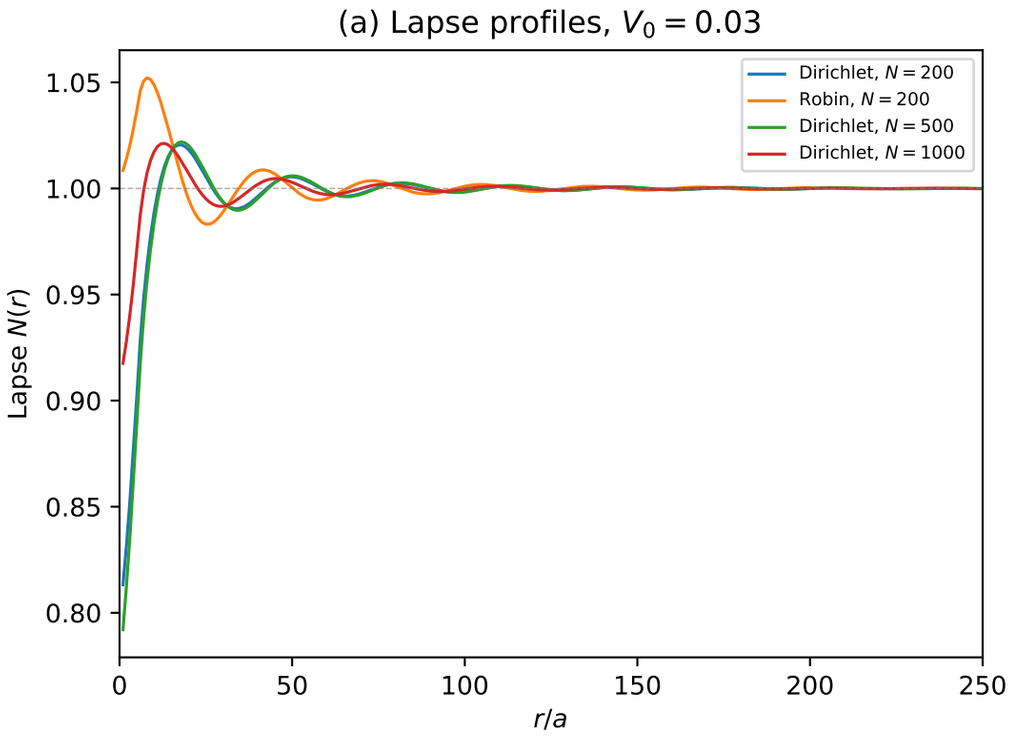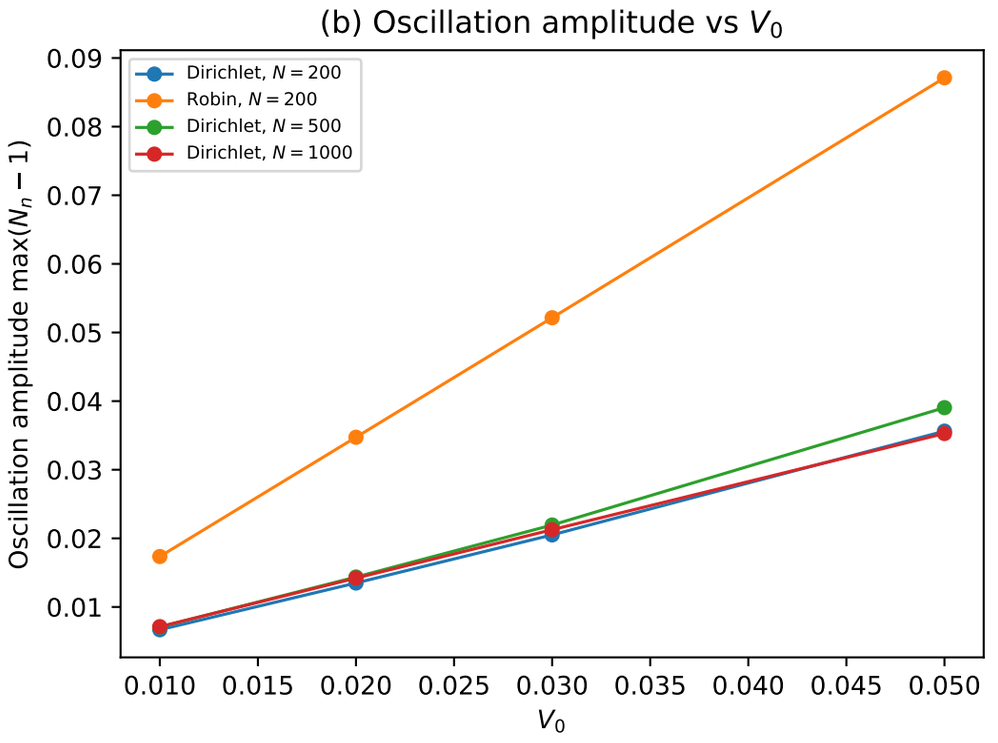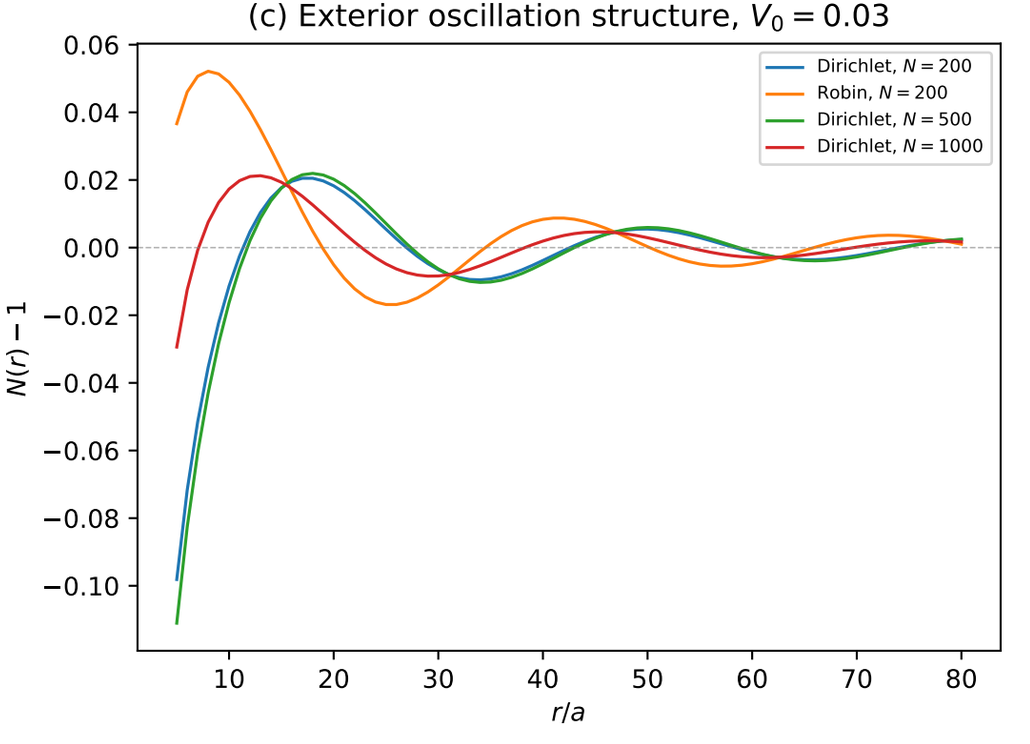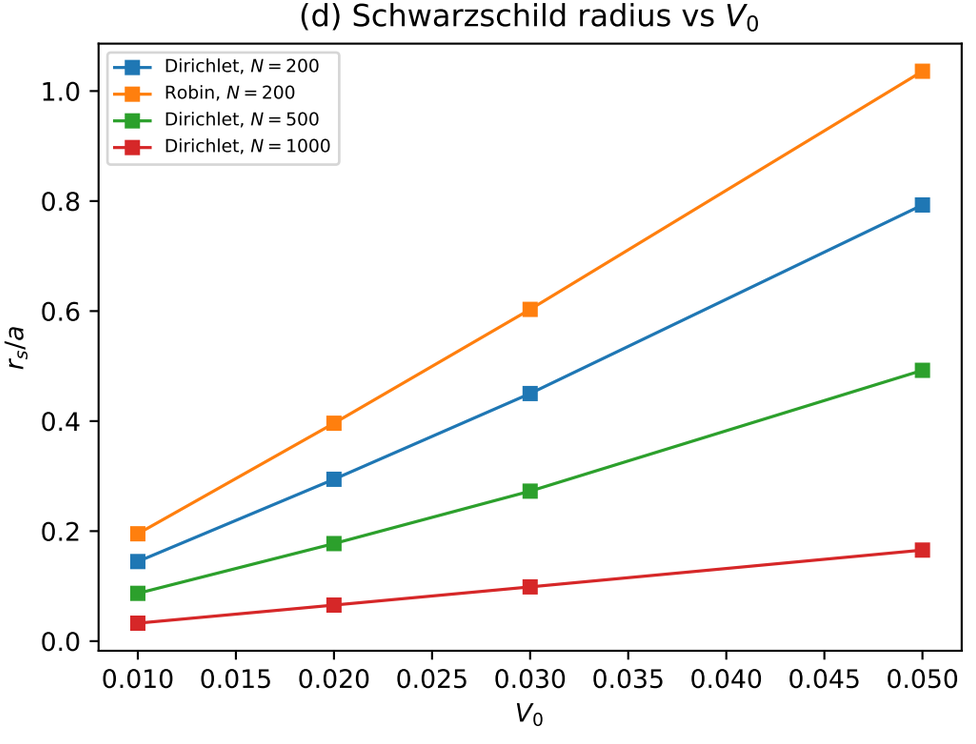Boundary Condition Study: Two-State Closure Equation